

Digital Signatures

Concepts, RSA and ElGamal Schemes

July 15, 2026

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RSA Digital Signature

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Introduction

Introduction to Digital Signatures

- A **digital signature** is a cryptographic mechanism used to verify the **authenticity** and **integrity** of a digital message or document.
- It is the electronic analogue of a handwritten signature, but far more secure.
- Based on **public-key (asymmetric) cryptography**: a *private key* to sign, a *public key* to verify.
- Provides three key security properties:
 - **Authentication** – confirms the identity of the sender.
 - **Integrity** – ensures the message was not altered.
 - **Non-repudiation** – sender cannot deny having signed the message.

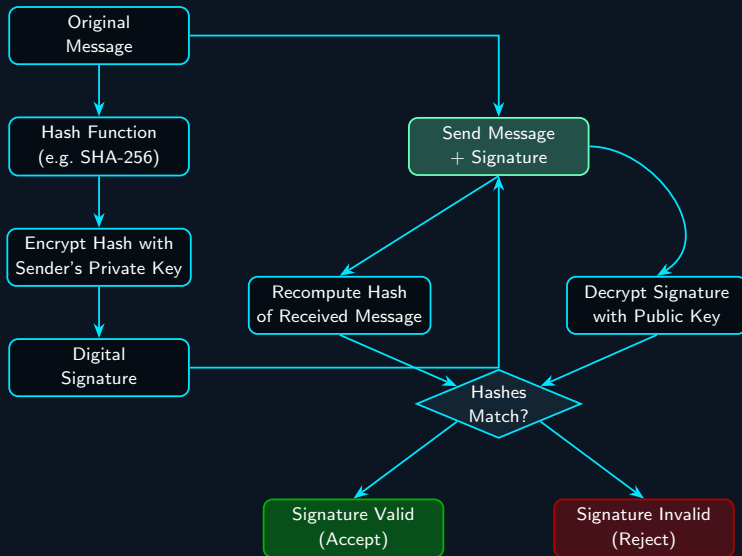
Applications

Applications of Digital Signatures

- Secure email (S/MIME, PGP)
- Software / code signing
- Financial transactions & e-banking
- Legal contracts and e-governance documents
- SSL/TLS certificates for websites
- Blockchain and cryptocurrency transactions
- Electronic tax filing
- Document management systems (PDF signing)

General Flowchart

General Flowchart of Digital Signature Process



Workflow

Workflow: How Digital Signatures Work

1. Key Generation

- Each user generates a pair of mathematically linked keys: a private key (kept secret) and a public key (shared openly).

2. Signing (by Sender)

- Compute a hash (message digest) of the document.
- Encrypt the hash using the sender's *private key* → this is the digital signature.
- Send the original message along with the signature.

3. Verification (by Receiver)

- Recompute the hash of the received message.
- Decrypt the received signature using the sender's *public key*.
- If both hash values match $\Rightarrow$ signature is **valid**; otherwise **invalid / tampered**.

RSA Digital Signature

RSA Digital Signature Scheme

Key Generation

1. Choose two large primes p, q . Compute $n = pq$.
2. Compute $\varphi(n) = (p - 1)(q - 1)$.
3. Choose public exponent e such that $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.
4. Compute private exponent $d \equiv e^{-1} \pmod{\varphi(n)}$.
5. Public key: (n, e) Private key: (n, d) .

Signing: $s \equiv m^d \pmod{n}$

Verification: accept if $m \equiv s^e \pmod{n}$

RSA Digital Signature – Small Example

Key Generation

$$p = 5, \quad q = 11 \quad \Rightarrow \quad n = pq = 55$$

$$\varphi(n) = (p - 1)(q - 1) = 4 \times 10 = 40$$

$$e = 3 \quad (\gcd(3, 40) = 1)$$

$$d = e^{-1} \bmod 40 = 27 \quad (\text{since } 3 \times 27 = 81 \equiv 1 \bmod 40)$$

Public key $(n, e) = (55, 3)$, Private key $(n, d) = (55, 27)$

Signing message $m = 2$:

$$s \equiv m^d \pmod{n} = 2^{27} \bmod 55 = 18$$

Verification:

$$s^e \bmod n = 18^3 \bmod 55 = 2 = m \quad \checkmark \text{ Signature valid!}$$

ElGamal Digital Signature

ElGamal Digital Signature Scheme

Key Generation

1. Choose a large prime p and a generator g of $\mathbb{Z}_p^*$.
2. Choose private key x , $1 < x < p - 1$.
3. Compute public key $y \equiv g^x \pmod{p}$.
4. Public key: (p, g, y) Private key: x .

Signing message m :

1. Choose random k with $\gcd(k, p - 1) = 1$.
2. Compute $r \equiv g^k \pmod{p}$.
3. Compute $s \equiv k^{-1}(m - xr) \pmod{p - 1}$.
4. Signature: (r, s) .

Verification: accept if $g^m \equiv y^r r^s \pmod{p}$

ElGamal Digital Signature – Small Example

Key Generation

$$p = 23, \quad g = 5, \quad x = 6 \text{ (private)}$$

$$y \equiv g^x \bmod p = 5^6 \bmod 23 = 8 \text{ (public)}$$

Public key $(p, g, y) = (23, 5, 8)$, Private key $x = 6$

Signing message $m = 10$, choose $k = 3$ ($\gcd(3, 22) = 1$):

$$r \equiv g^k \bmod p = 5^3 \bmod 23 = 10$$

$$s \equiv k^{-1}(m - xr) \bmod (p - 1) = 15 \times (10 - 60) \bmod 22 = 20$$

Signature: $(r, s) = (10, 20)$

Verification:

$$g^m \bmod p = 5^{10} \bmod 23 = 9, \quad y^r r^s \bmod p = 8^{10} \times 10^{20} \bmod 23 = 9$$

Both sides equal 9 ✓ Signature valid!